

120 Days of Quant Finance

Month 2: Derivatives & Time Series

Day 34: Value at Risk and Expected Shortfall

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Introduction

Measuring downside risk is central to financial risk management. Traditional metrics such as variance fail to capture the severity of extreme losses, especially for portfolios with nonlinear payoffs such as options.

Value at Risk (VaR) and Expected Shortfall (ES) are widely used risk measures designed to quantify potential losses over a specified time horizon and confidence level.

Value at Risk

Value at Risk measures the maximum expected loss over a given time horizon at a specified confidence level.

Let L denote the random variable representing portfolio losses. The Value at Risk at confidence level α is defined as:

$$\text{VaR}_\alpha = \inf \{l \in \mathbb{R} : P(L \leq l) \geq \alpha\}.$$

VaR corresponds to the α -quantile of the loss distribution.

Interpretation of VaR

A VaR of \$1 million at the 95% confidence level indicates that losses are expected to exceed \$1 million on only 5% of trading days. VaR does not provide information about the magnitude of losses beyond this threshold.

Limitations of Value at Risk

While VaR is simple and intuitive, it has several limitations. VaR does not capture tail severity, may fail to satisfy subadditivity, and can underestimate risk for portfolios with fat-tailed or skewed loss distributions.

Expected Shortfall

Expected Shortfall addresses the limitations of VaR by focusing on the average loss in the tail of the loss distribution.

Expected Shortfall at confidence level α is defined as:

$$ES_\alpha = E[L \mid L \geq \text{VaR}_\alpha].$$

Expected Shortfall provides information about the severity of losses when the VaR threshold is exceeded.

Comparison Between VaR and Expected Shortfall

Expected Shortfall is a coherent risk measure that satisfies subadditivity, making it more suitable for portfolios with nonlinear risk exposures.

For this reason, Expected Shortfall has been adopted as the preferred risk measure under modern regulatory frameworks such as Basel III.

Application to Option Portfolios

Option portfolios often exhibit asymmetric and fat-tailed loss distributions. VaR may underestimate the true risk of such portfolios, while Expected Shortfall captures the impact of extreme market movements more effectively.

Conclusion

Value at Risk and Expected Shortfall are fundamental tools in risk management. While VaR provides a useful threshold measure, Expected Shortfall offers a more complete view of tail risk. Together, these measures play a crucial role in the evaluation and control of financial risk.